CodeLlama 4309: natural-text expression

50 regressions: original base output passes; original merged output fails. Joint sparse latent measured on each text.

Measurement	Base-generated text	Merged-generated text
Tasks with any activation	50/50	44/50
Mean task-mean activation	0.1997	0.1979
Median first activation (active texts only)	31.5%	58.9%
First activation after main-function end	0/50	0/50

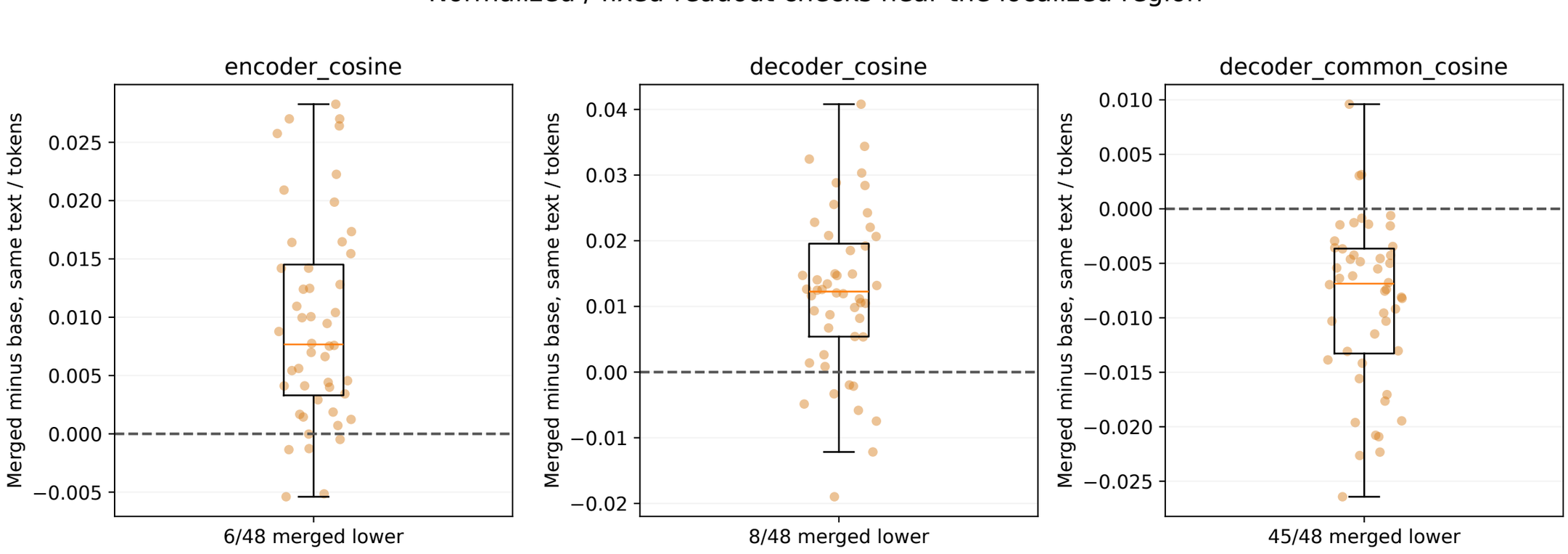
The shared latent is NOT generally stronger on failing merged texts by task-mean activation.
First-position percentages compare different generated programs and only active cases.
After-function excludes the newline ending the function itself; a return-line newline is not after-function.

Same text: base versus merged encoder contributions

Both models replay exactly the same original failing merged text. Contributions precede the JOINT ReLU / TopK.

Region	Base contribution	Merged contribution	Merged lower
Whole evaluated code	-0.999	-0.428	2/50
10 tokens before localized region start	-0.791	-0.210	12/48
Region start + next 10 tokens	-1.238	-0.584	5/48
Region start +/-10 tokens	-1.125	-0.504	4/48

Anchor is NOT the uniform DeepSeek test/import marker. It reuses existing failure localization:
16 runtime execution-site cases, 18 provisional line/interval cases and 16 omission regions.
Each anchor is transferred to frozen source code only via an exact matching text block.
Windows use available tokens; counts may differ by scope. These are NOT independent sparse activations.


Normalized / fixed-readout checks near the localized region
encoder_cosine
decoder_cosine
decoder_common_cosine
Merged minus base, same text / tokens
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
6/48 merged lower
8/48 merged lower
45/48 merged lower
Own encoder / own decoder directions, then one common decoder direction for both models.
Geometric readouts are not calibrated independent feature activations; selected exploratory cohort, no mediation claim.